

Prevalence and Nucleotide sequence relatedness among *Wenyonia* species (Cestoda: Caryophyllidea) in *Synodontis clarias* from Lekki Lagoon, Lagos, Nigeria

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Abstract

Background: *Wenyonia* Woodland, 1923 represents a monophyletic lineage of Caryophyllidean tapeworms that exclusively infect mochokid catfishes of the genus *Synodontis*. In Nigeria, information on the molecular characterization of *Wenyonia* sp tracing the phylogenetic relationship within the genus is lacking. This study was therefore aimed to evaluate the genetic relatedness between and among *Wenyonia* species based on the partial nucleotide sequence of the D1-D3 region of the 28SrRNA gene and the mt region (trnK + nad3 + trnS + trnW + cox1) genes.

Methods and results: One hundred and twenty *Synodontis clarias* from the Lekki Lagoon were purchased fresh at the Oluwo market Epe, Lagos, Nigeria between November 2020 and April 2021. Total genomic DNA of identified *Wenyonia* species recovered from the fishes were extracted. PCR was done to amplify the divergent domains (D1-D3) region of the 28SrRNA gene and the mt region (trnK + nad3 + trnS + trnW + cox1) genes in the *Wenyonia* species, and PCR products sequenced. The nucleotide sequences obtained were verified using BLAST and they were subsequently analyzed using the Molecular Evolutionary Genetic Analysis (MEGA 7.0). *Wenyonia* sp are abundant in *Synodontis clarias* with *W. virilis* the most prevalent and *W. minuta* the least prevalent. The phylogenetic tree based on the D1-D3 region of the partial 28SrRNA gene nucleotide sequence revealed three main clusters that indicated that *W. minuta* and *W. youdeowei* are more genetically related than *W. virilis*. *W. longicauda*, however, showed more intermediate relatedness between and among the 5 *Wenyonia* species used in the tree. *W. acuminata* showed the least genetically relatedness among the 5 *Wenyonia* species. Based on the partial mt region comprising trnK+ nad3+ trnS+ trnW+ cox1, *W. acuminata* showed more genetic relatedness with *W. longicauda* than with *W. youdeowei*.

Conclusion: The study therefore revealed that *Wenyonia* sp are abundant in *Synodontis clarias* and that the species are genetically closely related as revealed by the taxonomical classification. However, some individual among these species may have been evolving.

1.0. Introduction

Fish is a cheap source of animal protein with lots of dietary and health benefits. Freshwater fish such as the *Synodontis clarias* can serve as definitive, intermediate, or paratenic hosts in the lifecycle of many species of helminth parasites such as *Caryophyllideans* (Osimen and Anagba, 2020).

The *Caryophyllaeidea cestodes* are a group of unique parasitic worms that infect Siluriformes and Cypriniforms fishes causing diseases in the fish hence, negatively affect the fish's performance and yield. Woodland (1923) and (1937) created the genus *Wenyonia* since they differ from all other *Caryophyllaeidea cestodes*. The genus possesses sexual apertures in the anterior half of the body and a uterus whose longitudinal extent is almost equal to that of the testes. Ukoli, (1968) on the survey of the parasitic fauna of fishes in the River Niger, Shagunu, discovered some *Caryophyllaeidea cestodes* found in the intestine of the Siluriformes fishes *Synodontis* sp. He concluded that those cestodes belong to the

genus *Wenyonia* Woodland 1923, of which five species were recognizable and four of them belong to the genus *Wenyonia*. Ukoli (1968) thereafter reported the description of new species of *Wenyonia youdeowei*, *W. Synodontis*, *W. kainji*, and *W. mcconnelli*, all described from the intestine of *Synodontis* sp. *Wenyonia* is peculiar in that it has only been reported in the Siluriformes fishes and endemic to the freshwater systems of tropical Africa. Akinsanya and Otubanjo (2006) however, reported *Wenyonia acuminata* for the first time in *Clarias gariepinus* from Lekki Lagoon, Lagos, Nigeria and also confirm first record of *Paracamallanus cyathopharynx* in the same fish species. Akinsanya and Otubanjo (2006) and (2008) reported first record of *Wenyonia acuminata* and *Raphidas carodes* sp from the intestine of *Clarias gariepinus* and *Synodontis clarias* respectively and concluded that the occurrence of these parasites in the fish species may be indicative of similar diets and feeding habits amongst the freshwater fishes.

With the employment of nucleotide sequencing and other molecular analytical processes conflicting results between morphological and genetic data on the current taxonomy and systematics of a particular organism are emerging, showing or indicating incorrect taxonomic identification and grouping of certain species. Bazsalovicsová *et al.*, (2020) reported some examples, including Scholz *et al.*, 2011, a review on the key to the identification and molecular phylogeny

of *Khawia* sp. Parasites of cyprinid fish and Schaeffner *et al.*, (2011) on *Wenyonia* Woodland from catfishes in Africa:

the study showed that no genetic differences exist in the partial 18S rDNA of different morphotypes of *W. virilis*. Hanzelová *et al.*, (2015) reported conflicting findings between morphological and molecular data for *Caryophyllaeus* sp (*C. laticeps* and *C. chondrostomi*). The findings led to the identification of a different morphotype, which was somewhat distinct from most frequent “nominotypical” morphotype, hence, the description of the individuals of the morphotype as a new species, *C. chondrostomi*.

Molecular assays have therefore proven to offer better advantages and are suitable in the characterization of organisms and identification of individuals. Employment of both morphological/morphometric indexes and molecular methods presents the best approach in the taxonomical, systematic, and evolutionary characterization of organisms and or individuals (Bazsalovicsová *et al.*, 2020). In Nigeria, there is a dearth of information on the molecular characterization of *Wenyonia* sp tracing the phylogenetic relationship within the genus. It is, therefore, pertinent to investigate the genetic diversity which may exist within the *Wenyonia* group examined. This study was therefore aimed to evaluate the genetic relatedness between and among *Wenyonia* species based on the partial nucleotide sequence of the D1-D3 region of the 28S rRNA gene and the mt region (trnK + nad3 + trnS + trnW + cox1) genes.

2.0 Material And Methods

2.1. Ethical clearance

Ethical clearance for the study with the reference number CMUL/ACUREC/03/20/730 was gotten from the health research ethics committee of the College of Medicine, University of Lagos, Nigeria.

2.2. Fish host sampling

Eighty *Synodontis clarias* from the Lekki Lagoon (6°25'- 6°35'N and 4°00'- 4°13'E) were purchased fresh at the Oluwo market Epe, Lagos, Nigeria between January 2021 and April 2021. The fishes were rendered inactive by cervical dislocation for easy handling prior to dissection.

2.3. Collection and identification of parasites

The fishes were dissected, the intestines were removed and placed in Petri dishes containing normal saline. The intestines were carefully slit open to aid the emergence of parasites. A total of 897 helminths parasites were recovered from all the fish used in this study following Akinsanya (2015). The collected helminths parasites were counted, recorded, and fixed in 70% alcohol. Parasite identification was done using the keys provided by (Wooland, 1937, Yamaguti, 1959, Ukoli, 1972, and Paperna, 1996). The cestode parasites to be used for molecular studies were preserved in 95% ethanol until ready for use.

2.4. Genomic DNA extraction

Total genomic DNA of representatives of each *Wenyonia* sp were extracted using the Jena bioscience Blood-Animal-Plant DNA preparation columns kit. This was done following the manufacturer's instructions. In brief: 300 µl Lysis Buffer and 2 µl RNase A were added to each parasite and vortex vigorously for 30–60 sec. 8 µl Proteinase K was added, and incubation was done at 60°C for 20 min and allowed to cool down for 5 min. 300 µl Binding Buffer was added, and vortex briefly. The tube was then placed on ice for 5 min and centrifuge for 5 min at 10,000 g. The supernatant was pipetted directly into the spin column, centrifuge for 1 min at 10,000 g, while the flow-through was discarded. 500 µl Washing Buffer was added into a spin column, centrifuge for 30 sec at 10,000 g, and the flow-through was discarded. 500 µl Washing Buffer was added, for a second wash, into the spin column, centrifuge for 30 sec at 10,000 g, and the

flow-through discarded. It was centrifuged again at 10,000 g for 1 min to remove residual Washing Buffer. The 2ml wash tube was discarded, and the column was placed into the elution tube. 50 µl Elution Buffer was added into the center of the column, incubated at room temperature for 1 min, and then centrifuged at 10,000 g for 2 min. The DNA was then stored at 4°C.

2.5. Gene amplification (polymerase chain reaction)

PCR to amplify the divergent domains (D1-D3) region of the 28SrRNA gene and the mt region (trnK + nad3 + trnS + trnW + cox1) genes in the *Wenyonia* species was performed in a total volume of 25µl reactions consisting 5µl of DNA template, 1µl each of the primers, 5µl of 5X FIREPol master mix (Solis BioDyne), and 13µl of double-distilled water. The amplification was performed using the primers and cycling conditions shown in Table 1.

2.6. Gel electrophoresis

Amplified PCR products were analyzed with 2% ethidium- bromide-stained agarose gel electrophoresis at 90V for 45mins then viewed under UV light.

2.7. Nucleotide sequencing and sequence analysis

The PCR products were sequenced using the ABI 3500XL Genetic Analyzer (Inqaba Biotech South Africa). The nucleotide sequence obtained was verified using BLAST to confirm they were D1-D3 region of 28SrRNA and mt region comprising trnK + nad3 + trnS + trnW + cox1 genes. The sequences of reference species were retrieved from GenBank (www.ncbi.nlm.nih.gov) as presented in Table 2. The nucleotide sequences were subsequently analyzed using the Molecular Evolutionary Genetic Analysis (MEGA 7.0). The evolutionary history was inferred using the Neighbor-Joining method. The optimal tree with the sum of branch length = 20.45618905 is shown. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (100 replicates) are shown next to the branches. The phylogenetic tree was drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The maximum Composite Likelihood method was used to compute the evolutionary distances and is in the units of the number of base substitutions per site. Evolutionary analyses were conducted in MEGA7.0.

Table 1
Primer sequences and PCR cycling conditions used for genes amplifications

Gene	Primer	Sequence (5'-3')	Cycling conditions
D1-D3 region of 28SrRNA	LSU- 5	TAGGTCGACCCGCTGAAYTTAAGCA	Initial denaturation for 5min at 95°C, followed by 35 cycles of 30sec at 95°C, 30sec at 57°C, 1.5min at 72°C, and completed by 10min extension hold at 72°C
	1500R	GCTATCCTGAGGGAAACTTCG	
		(Littlewood <i>et al.</i> , 2000; Olson <i>et al.</i> , 2003;	
		Schaeffner <i>et al.</i> , 2011)	
mt region comprising trnK + nad3 + trnS + trnW + cox1	CFCYT1	GCAGGTTACTTTGATATAG	Initial denaturation for 5min at 95°C, followed by 30 cycles of 1min at 94°C, 1min at 50°C, 2min at 72°C, and completed by 5min extension hold at 72°C
	CRCYT2	CCAAAAAACC AAAACAT	
		(Bazsalovicsova <i>et al.</i> , 2011; Brabec <i>et al.</i> , 2012)	

Table 2
Referenced *Wenyonia* sp nucleotide sequences accession numbers from Gene Bank

Gene	<i>Wenyonia</i> species	Accession number
28SrRNA (partial sequence)	<i>Wenyonia youdeowei</i>	HQ848491.1
28SrRNA (partial sequence)	<i>Wenyonia acuminata</i>	HQ848489.1
28SrRNA (partial sequence)	<i>Wenyonia virilis</i>	HQ848522.1
28SrRNA (partial sequence)	<i>Wenyonia minuta</i>	HQ848507.1
28SrRNA (partial sequence)	<i>Wenyonia youdeowei</i>	HQ848496.1
28SrRNA (partial sequence)	<i>Wenyonia minuta</i>	HQ848518.1
28SrRNA (partial sequence)	<i>Wenyonia virilis</i>	HQ848521.1
28SrRNA (partial sequence)	<i>Wenyonia acuminata</i>	MW027418.1

3.0. Results

A total of 642 *Wenyonia* sp (*W. virilis*, *W. acuminata*, *W. youdeowei*, *W. longicauda*, *W. synodontis*, and *W. minuta*) were identified and recorded as shown in (Fig. 1) below.

The genetic divergence and phylogenetic relatedness among *Weynonia* sp present in the intestines of the *Synodontis clarias* from the Lekki Lagoon based on the partial gene sequences of the D1-D3 region of 28SrRNA and the mt region comprising trnK + nad3 + trnS + trnW + cox1 is presented here. Phylogenetic relations estimated by Maximum Composite likelihood and the Neighbor-Joining methods resulted in the trees with identical topologies, as shown in (Figs. 2–4). The phylogenetic tree based on the D1-D3 region of the partial 28SrRNA gene nucleotide sequence revealed three main clusters (Fig. 2). The clustering patterns indicated that *W. minuta* and *W. youdeowei* are more genetically related than with *W. virilis*, which was distantly related in a separated cluster. However, *W. longicauda* showed more intermediate relatedness between and among the 5 *Wenyonia* species used in the tree. *W. acuminata* showed the least genetically relatedness to the other four *Wenyonia* species.

Based on the partial mt region comprising trnK + nad3 + trnS + trnW + cox1, *W. acuminata* showed more genetic relatedness with *W. longicauda* than with *W. youdeowei* (Fig. 3). The nucleotide sequence reads of *W. virilis* and *W. minuta* generated could not be used for the phylogenetic tree.

Three (3) main clusters, with some sub-clusters, were generated based on the nucleotide sequences of the D1-D3 region of 28SrRNA gene partial nucleotide sequences of the studied population and that of the selected reference sequences (Fig. 4). The referenced sequences were mostly clustered within the first cluster separated from the studied population in the second and third clusters. *W. virilis* and *W. minuta* showed the least genetic relatedness amongst the 5 *Wenyonia* species been studied. The second cluster

consists of *W. acuminata*, *W. longicauda*, *W. minuta*, and *W. youdeowei*, while the third cluster was made up of *W. virilis*, *W. longicauda*, *W. youdeowei*, and *W. acuminata*.

4.0. Discussion

Wenyonia species are important parasites of the catfish, influencing the overall performance of the fish, which in turn affects its economic value contributes to global food security. Diversity in morphological features of these fish parasites has been documented; however, the genetic and evolutionary relatedness amongst the species is lacking. The present study instigated the genetic relatedness among the species and revealed that a certain degree of the evolutionary process might have occurred among the five *Wenyonia* sp present in the intestines of the *Synodontis clarias* from the Lekki Lagoon.

Some levels of species interrelatedness were seen among the *Wenyonia* species: closest genetic relatedness was shown by *W. minuta* and *W. youdeowei*. *W. virilis* was shown to be least genetically related among the species. *W. longicauda* was, however, seen to show some level of intermediated relatedness between and among the five species based on the D1-D3 region of the 28S rRNA gene. This observed interrelatedness may indicate that the gene may have been evolving within and among these species. Therefore, the phylogenetic tree places *W. longicauda* in a position that shows this species to be the most genetically related to all the other four *Wenyonia* species investigated. Schaeffner *et al.*, (2011) phylogenic analysis based on the partial 28S rRNA gene sequences grouped the *Wenyonia* species; *W. acuminata*, *W. minuta*, *W. virilis*, and *W. youdeowei*, into two lineages: close relatedness between *W. acuminata* and *W. minuta* in a clade, and *W. virilis* and *W. youdeowei* in another clade. The divergent domains (domains D) of the 28S ribosomal RNA gene have been confirmed to be a good marker for molecular classification and identification of some groups of parasites (Zhao *et al.*, 2020). Therefore, the relatedness as shown by the species as revealed by the amplified genes' segment provided some insight into a possible molecular classification of the individual species.

Based on the partial mt region, *W. acuminata* is more genetically related to *W. longicauda* than *W. youdeowei*. Jirsova *et al.*, (2017) revealed that mtDNA analysis showed that the *Wenyonia youdeowei* and *W. minuta* did not

separate from *W. virilis*. In the present study, although members of each species did not entirely separate into different clusters or clades, each species was distinguished from the other.

The phylogenetic analysis separated the studied *Wenyonia* species population from the reference sequences, indicating that some environmental factors may play a significant role. Rapid evolution process in the parasites has been reported (Jirsova *et al.*, 2017). Several evolutionary factors may be involved. The reports suggested that host's higher migration rates of the host, differences in effective population sizes, species-specific historical events, or a higher life expectancy of the host, may play a role to increase the mutation rate in the parasite (Kaltz and Shykoff 1998; Gandon and Michalakis 2002; Jirsova *et al.*, (2017). Jirsova *et al.*, (2017), in their report, further suggested that host-parasite coevolution may be taking place as well. This shows that the *Wenyonia* sp in the population investigated may have

been evolving and however, they are still a genetic line (genus) as revealed by the taxonomical classification. It is therefore, of importance that a deeper and larger genetic investigation is done to ascertain species variations or differences based the morphological and or taxonomical classification or difference.

The lack of molecular data caused a limitation in phylogenetic relationship studies of *Wenyonia* species. In the present study, we successfully screened evaluated divergent regions of 28S rDNA and mt region (trnK + nad3 + trnS + trnW + cox1) genes among *Weynonia* species based on bioinformatics analysis as generated using the genes partial nucleotide sequences of the sampled population and those retrieved from GenBank, towards affirming the relatedness between and among these species.

Therefore, it is of great significance that molecular approaches are employed in characterizing these species combined with the morphological taxonomy and systematic classifications for more accurate identification.

Declarations

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Author Contributions

Conceptualization- Akinsanya, Bamidele, and Adekoya, Khalid Olajide

Investigation- Koledoye, Toluwani Yemi, and Abazuh, Uchenna Desmond

Resources- Koledoye, Toluwani Yemi

Writing original draft- Koledoye, Toluwani Yemi, and Abazuh, Uchenna Desmond

Review and Editing- Abazuh, Uchenna Desmond, Koledoye, Toluwani Yemi, Akinsanya, Bamidele and Adekoya, Khalid Olajide

All the authors read and approved the final article.

Ethics approval

Ethical clearance for the study with the reference number CMUL/ACUREC/03/20/730 was gotten from the health research ethics committee of the College of Medicine, University of Lagos, Nigeria.

Consent to participate: N/A

Consent to publish: N/A

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Figures

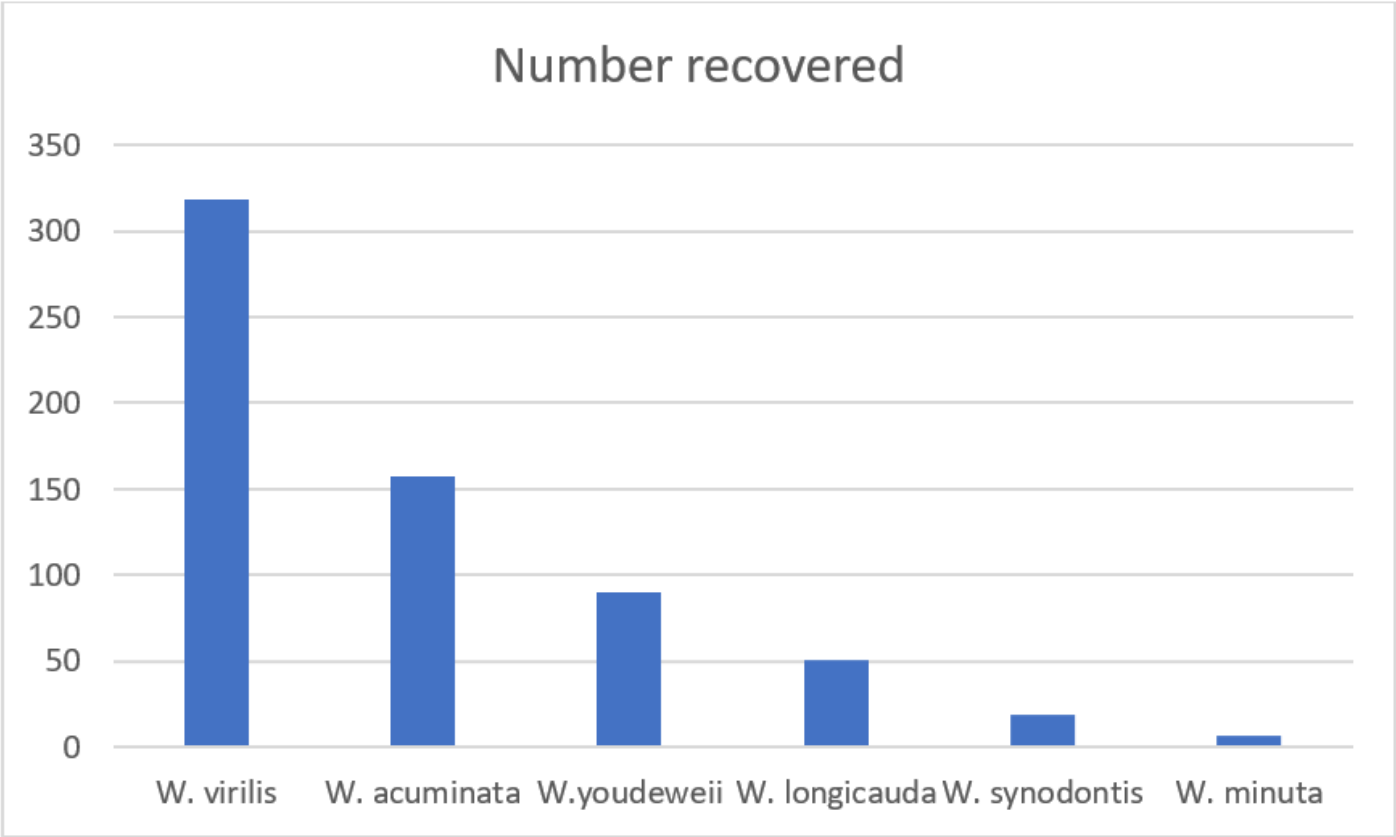


Figure 1

Bar chart showing the different Wenyonina species recovered from Synodontis clarias from Lekki Lagoon.

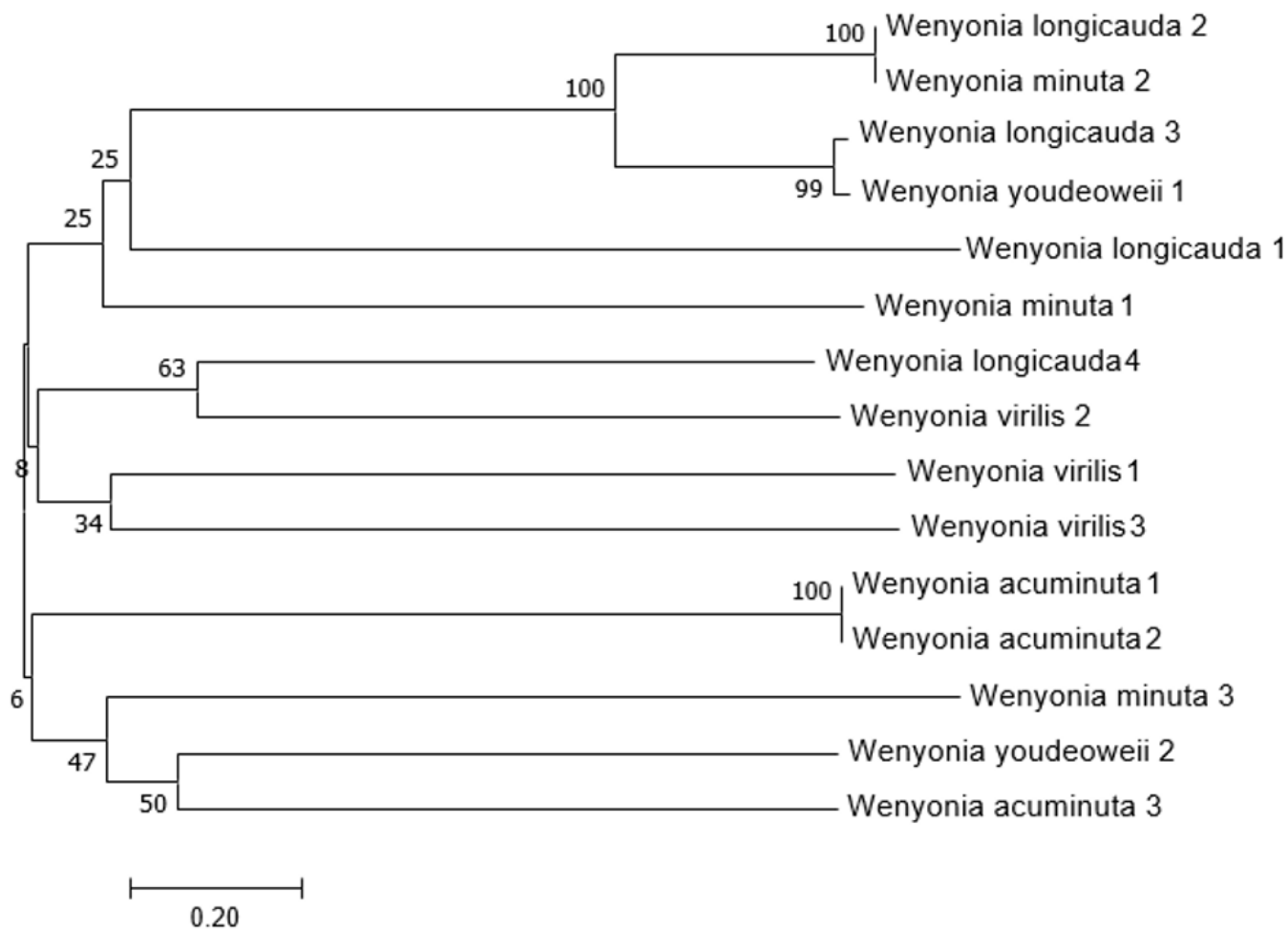


Figure 2

Phylogenetic tree based on nucleotide sequence in the D1-D3 region of 28SrRNA gene.

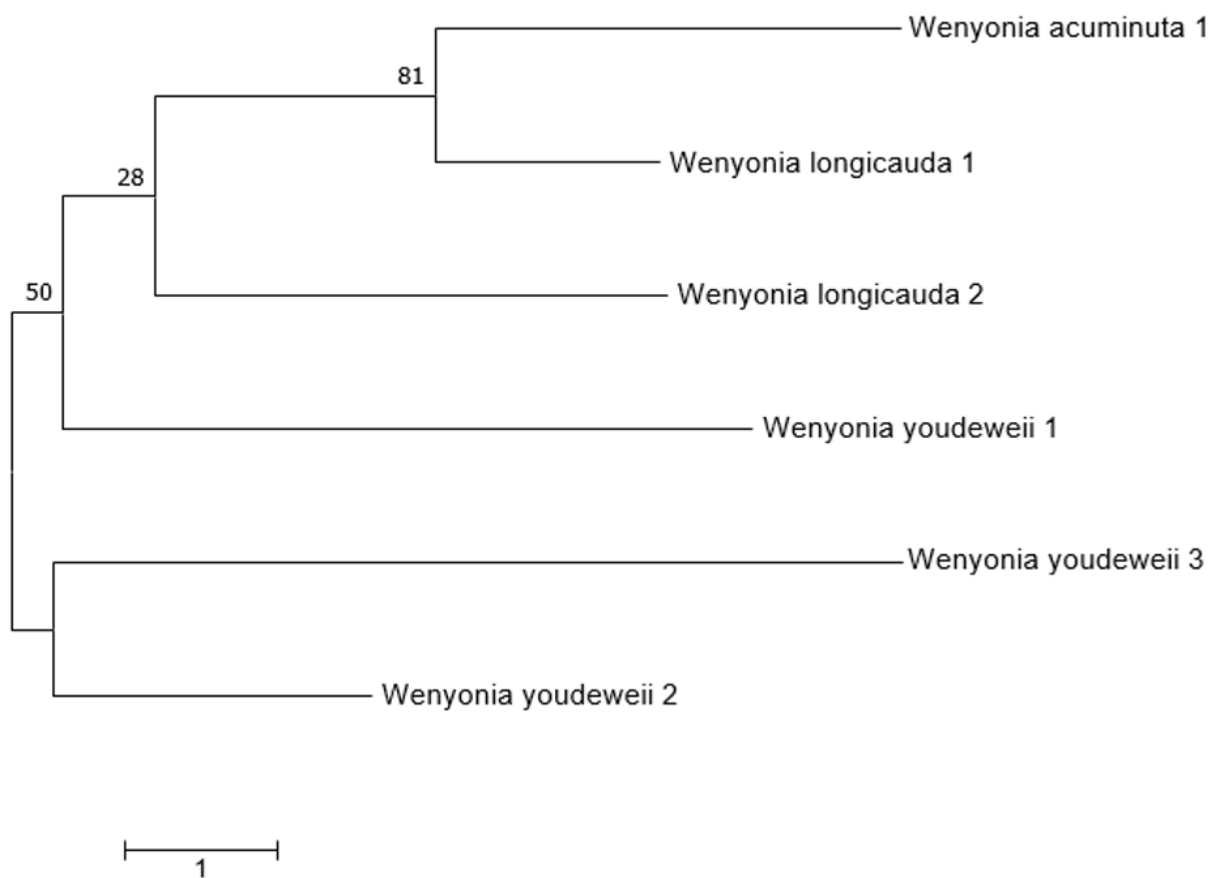


Figure 3

Phylogenetic tree based on nucleotide sequence in the mt region comprising trnK+ nad3+ trnS+ trnW+ cox1.

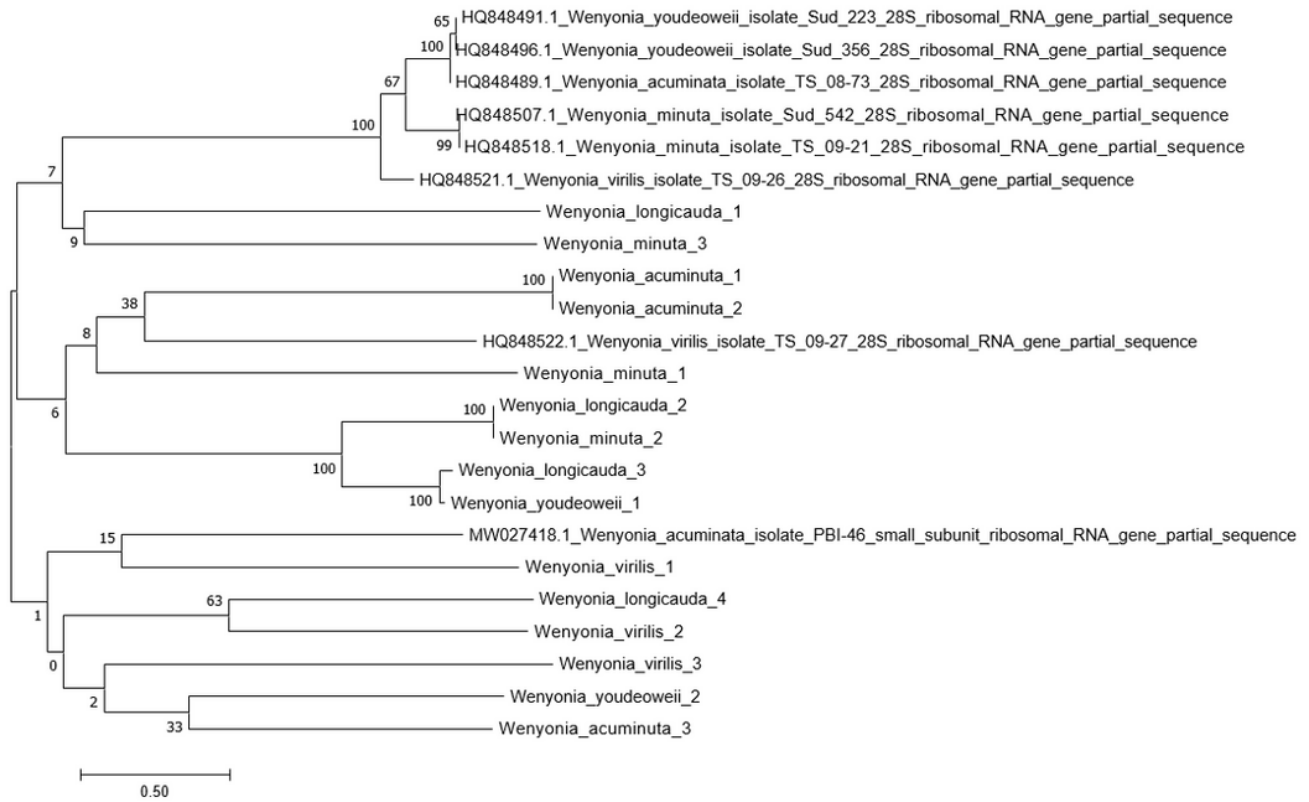


Figure 4

Phylogenetic tree with reference RNA based on the D1-D3 region of 28SrRNA gene